

External validation of a two-timescale adaptive-margin model of background-conditioned physiological vulnerability

(authors)

June 13, 2026

Abstract

The TwoTimescaleResilience framework models *background-conditioned vulnerability*: chronic environmental pressure and retained chemical burden slowly narrow a species' adaptive margin A_t , weakening its restoring force $\lambda(A_t)$ and amplifying the burden of a later acute perturbation. Capacity parameters are derived offline from Add-my-Pet (AmP/DEB) records, pressure from EPA ECOTOX, and memory from a compound-retention recurrence. Here we confront the recovery/margin layer with *independent* external data at multiple organisational levels, using rank statistics and pre-registered effect signs throughout. (i) At the **population** level, the DEB maintenance rate constant k_M predicts demographic recovery (damping ratio) in the COMADRE matrix database beyond pace-of-life and a real dated phylogeny ($\rho=+0.22$, $p=0.011$, rank PGLS), and the DEB reproduction rate R_i predicts demographic compensation ($\rho=+0.77$). (ii) At the **individual** level, the modelled margin tracks Scope for Growth where tissue burden indexes exposure (Widdows 1995, $\rho=+0.41$), survival-in-air under emersion (ICES DOME, $\rho=+0.39 \rightarrow +0.45$ under body-size and condition control), and a controlled dose-response and additive mixture (Viarengo 1995). (iii) The erosion **dynamics** reproduce continued multi-month decline that a static map cannot (transplant proof-of-concept). The one-dimensional amplification scalar $g/\mathcal{F}$ is **null in every test**, as the margin-first reading predicts. Two well-powered **negative controls** bound the claim: the single-trait link from k_M to acute toxic sensitivity ($n=310$) is fully accounted for by body size, so the model's distinctive leverage is its across-axis *capacity weighting*, not k_M alone. Effects are rank-robust, magnitude-modest, and specification-sensitive—corroboration of the margin/recovery layer, honestly bounded.

1 Introduction

The framework and its mathematics are developed elsewhere (the companion paper; also Kooijman, 2010). In brief, for a target species it maps environmental pressure to burdens on four DEB physiological axes (assimilation, maintenance, growth, reproduction), aggregates these into a scalar *adaptive margin* A_t , and lets A_t set a slow restoring force $\lambda(A_t)$ whose two-timescale interplay with acute perturbation defines vulnerability. The capacity layer is parameterised offline from AmP (Marques et al., 2018), pressure from ECOTOX (Olker et al., 2022), and memory from a compound-retention recurrence.

A model assembled from independent ingredients is testable without circularity provided the *outcome* side never uses a model-derived quantity. We therefore confront the framework with external data at several organisational levels, asking a deliberately conservative question at each: does a model quantity predict an independently measured outcome, *beyond* the obvious confounds (pace-of-life, phylogeny, body size)? Three commitments keep the test honest: (a) **rank statistics** throughout (unit-invariant, robust to the monotone-but-nonlinear responses the model produces); (b) **pre-registered effect signs**; and (c) explicit reporting of nulls and bounds.

A central interpretive point—the *margin-first* reading—is that the validated product is the margin/recovery layer (its rate endpoints, state, function, and dynamics), *not* the one-dimensional amplification scalar $\mathcal{F} = \lambda(A_0)/\lambda(A_t)$, which by construction can carry only one physiological number per species and which we find predicts nothing externally.

2 Methods

Static (point) response. Every cross-sectional test uses the instantaneous map `compute_adaptive_margin_response` per-axis pressure x_i (a threshold-free, median-normalised relative tissue burden, $x_i = c_i/\text{median}(c_i)$) becomes a bounded impairment $E_i = x_i/(1 + x_i)$; impairments aggregate as $Q = \sum_i w_i E_i$ with AmP-derived weights w_i ; and the margin is $A_t = A_0(1 - Q)$. This exercises the response-curve shape, the mode-of-action (MoA) axis routing, and the AmP capacity weighting, but not the time dynamics.

Dynamic response. The transplant test integrates the slow erosion state, which relaxes at rate $\lambda \in [k_M, k_M g]$ with time-constant $1/\lambda$; for *M. edulis* $1/\lambda \approx$ months, *unfitted* (it falls out of the maintenance rate $k_M = [p_M]/[E_G]$).

MoA routing. Contaminants route to axes by physiological mode of action: hydrocarbons/PAH $\rightarrow$ assimilation; metals $\rightarrow$ maintenance (a deliberate confound-control for field tissue burden; see §4); organotin/organochlorine $\rightarrow$ reproduction; dibutyltin $\rightarrow$ growth.

Controls and statistics. Recovery endpoints are controlled for pace-of-life (generation time) and phylogeny (taxonomic-Order proxy and a dated-tree PGLS with Pagel’s λ by maximum likelihood; Pagel, 1999); individual-level energetics are controlled for body size and condition. All correlations are Spearman rank coefficients; $*p < 0.05$, $**p < 0.01$.

Data anchors. COMADRE (Salguero-Gómez et al., 2016); GlobTherm (Bennett et al., 2018); Scope for Growth (Widdows et al., 1995, 2002; Albentosa et al., 2012); survival-in-air (ICES DOME OSPAR CEMP, ICES, 2024) and controlled stress-on-stress (Viarengo et al., 1995); transplant and single-contaminant time-courses (Veldhuizen-Tsoerkan et al., 1991b,a); toxicokinetics (Rubach et al., 2010) and DEBtox/GUTS calibrations (Nyman et al., 2012; Ashauer et al., 2016; Baas and Kooijman, 2015; Jager, 2020; Jager et al., 2011).

3 Results

3.1 Recovery rate endpoints — COMADRE (population demography)

The DEB maintenance rate $\lambda_{\min} = k_M$ predicts the COMADRE log damping ratio (recovery rate) across $n=197$ matched species: raw $\rho = +0.41^{**}$, partial on generation time $+0.26^{**}$, and within-Order $+0.19^*$ — i.e. beyond pace and coarse phylogeny. A *dated* TimeTree (Kumar et al., 2017) PGLS over 182 species resolves the phylogeny question: Pagel’s $\lambda \approx 0$ (the trait carries little phylogenetic signal), and in rank form $k_M \rightarrow$ recovery **survives** the generation-time control under the dated covariance ($\beta^*=0.221$, $p=0.011$), while the log-linear form nulls ($p=0.96$). The effect is thus genuinely rank/monotone, robust to pace and real phylogeny simultaneously. The per-axis reproduction rate R_i predicts demographic *compensation* $\rho = +0.77^{**}$ (beyond pace and size) — the strongest single result. The amplification scalar g is null at every level (-0.05 to -0.13).

3.2 Capacity coherence and its bound — GlobTherm

In a pre-registered probe ($n=664$ AmP $\leftrightarrow$ GlobTherm species), the AmP capacity axis correlates with independent thermal-tolerance data ($|\rho|$ up to 0.45) — the offline AmP $\rightarrow$ capacity extraction carries real biological structure. But higher $k_M/\lambda_{\max}$ implies *narrower* thermal breadth (-0.45 all-taxa), **refuting** a general-resilience reading: recovery capacity is specific to demographic recovery, not a universal resilience currency.

3.3 The margin state — Scope for Growth

At the margin’s own organisational level (no scale bridge), the modelled margin tracks measured SFG along contaminant gradients in a scale-dependent way: $\rho = +0.41^*$ where tissue burden indexes exposure (Widdows 1995, estuary), $+0.12$ at basin scale (Widdows 2002), and -0.11 where condition/food dominates (Albentosa 2012; the authors find SFG $\sim$ condition -0.62^{***}). Across all three, metals behave as a *positive* confound (tissue burden tracks condition, not exposure); the MoA routing keeps them off the toxic axis, so the routed margin beats naive equal-weight load.

3.4 The margin function — stress-on-stress

Survival-in-air under emersion is the closest *outcome* to the margin’s purpose. In the ICES DOME panel (*M. edulis*, 17 UK stations), survival tracks the modelled margin $\rho = +0.39 \rightarrow +0.45$ under body-length and condition control. Decisively, and unlike Albentosa, *controlling* the condition confound here *strengthens* the signal — genuine margin erosion, not a health-proxy artifact. A controlled-exposure cross-check (Viarengo et al., 1995) gives a monotone dose-response (potency Cu>PAH>PCB) and a binary mixture that is additive with no antagonism, bracketed by the model’s own concentration-addition / independent-action rules.

3.5 The margin dynamics — transplant proof-of-concept

The only test that runs the erosion dynamics (Veldhuizen-Tsoerkan et al., 1991b): cadmium plateaus by 2.5 months yet survival keeps dropping to 5 months. A static map predicts no further erosion; the dynamic state rises $\sim 33\%$ on the model’s *unfitted* months-scale $1/\lambda$, matching the continued decline ($n=4$, proof-of-concept). A single-contaminant firm-up (Veldhuizen-Tsoerkan et al., 1991a) removes the PCB confound: cadmium alone erodes survival progressively, modelled margin tracking it $\rho = +0.90$.

3.6 The consistent null — the amplification scalar

Across *every* anchor the one-dimensional amplification scalar $g/\mathcal{F}$ predicts nothing. This is the margin-first prediction confirmed: external support is for the margin/recovery layer, not for a scalar amplification readout.

3.7 Bounding negative controls — single-trait $k_M \rightarrow$ toxicity is size-confounded

Two direct, out-of-sample tests of “does maintenance pace predict toxic response?” both reduce to a body-size confound. (i) *State axis*: species-level acute LC50/EC50 for four AChE inhibitors from EPA ECOTOX, matched to AmP ($n=310$; replicating Baas and Kooijman, 2015). Higher $k_M/[p_M] \rightarrow$ more sensitive ($\rho \approx -0.27$, all four chemicals) *raw*, but the partial controlling body size is ≈ -0.03 — a robust null at $n=310$. (ii) *Rate axis*: against measured chlorpyrifos elimination rate constants (Rubach et al., 2010), k_M is weak/wrong-signed and non-significant; and elimination is a

Table 1: Honest scorecard. Effects are Spearman ρ ; * $p < 0.05$, ** $p < 0.01$.

Anchor	Level	Result	
COMADRE (k_M , R_i)	population	$k_M \rightarrow \text{recovery} +0.19 \rightarrow +0.22^*$ (rank, dated PGLS); $R_i \rightarrow \text{comp.} +0.77^{**}$	c
GlobTherm ($n=664$)	physiology	capacity coherent ($ \rho \leq 0.45$), recovery-specific	b
Scope for Growth ($\times 3$)	energetics	$+0.41 \rightarrow +0.12 \rightarrow -0.11$ (scale-dependent)	c
Stress-on-stress (DOME)	energetics	$+0.39 \rightarrow +0.45$ (confound-controlled)	s
Viarengo 1995	controlled dose	monotone dose-response; additive mixture	c
Transplant (Veldhuizen)	dynamics, over time	continued erosion static map cannot; Cd-alone $+0.90$	p
$k_M \rightarrow \text{toxicity}$ ($n=310$)	cross-species	raw -0.27 ; nulls under body size	b
amplification $g/\mathcal{F}$	—	null everywhere	n

toxicokinetic, not the thesis-relevant toxicodynamic recovery rate (Nyman et al., 2012). The reading: the model’s distinctive leverage is its across-axis *capacity weighting*, not k_M as a scalar predictor.

4 Discussion

External evidence supports the **margin/recovery layer** at four organisational levels — its rate endpoints, its state, its function, and a first sign of its dynamics — while the amplification scalar is null throughout, exactly as the margin-first reframe predicts. Three honest qualifications define the result.

Rank-robust, magnitude-modest, specification-sensitive. Effects are $\rho \approx 0.2\text{--}0.45$ — corroboration, not strong prediction — and several live in the ranks (the headline k_M result survives a dated-tree PGLS in rank form but nulls in log-linear form). They should be reported as monotone tendencies.

Tissue burden $\neq$ exposure, and the metals lesson. The pressure proxy is a relative tissue burden; where burden tracks food/condition rather than exposure it fails (Albentosa). This is why metals route to maintenance in field data: it is a *confound-control*, not a mechanistic claim. Fitted DEBtox modes of action place metals on assimilation, but re-routing metals to assimilation *degrades every field anchor* (e.g. Widdows 1995 $+0.41 \rightarrow +0.30$; Widdows 2002 $+0.12 \rightarrow -0.14$). The two assignments therefore coexist by data regime: field measured burden uses the maintenance confound-control; controlled or exposure-derived burden may use the fitted assimilation mode.

The distinctive content is the capacity weighting, and it is carried as an assumption. Every individual-level test is single-species, so the AmP capacity weighting is held constant; and the single-trait $k_M \rightarrow \text{toxicity}$ link is body-size-confounded. The model’s distinctive across-axis capacity weighting therefore remains *untested*, awaiting across-species contaminant-gradient data that largely does not exist; it is stated as a model assumption, like the mixture rules.

Scope for water-quality application. The validated machinery (margin, saturating response, recovery dynamics) supports *relative*, mechanistically structured vulnerability statements — ranking sites and months under sustained pressure — rather than absolute risk thresholds or across-species predictions. That is the defensible use the present evidence licenses.

Reproducibility and data availability

All anchors use openly available data (COMADRE, GlobTherm, ICES DOME OSPAR CEMP, EPA ECOTOX, AmP) or published tables; raw downloads are gitignored and derived tables, harnesses, and analysis scripts are in the repository (`examples/`, `scripts/`, `data/external/`). A consolidated account with per-anchor detail is `docs/notes/external_validation_synthesis.md`. Statistics are Spearman rank correlations with pre-registered signs; analyses run under Julia 1.12.6.

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